

Grinold's (1989) "fundamental law" of active management makes this intuition formal. It states that the maximum information ratio attainable—since it ignores transactions costs, restrictions on trading, and other real-world considerations—is given by:

$$IR \approx IC \times \sqrt{BR}, \quad (10.5)$$

where IR is the information ratio; IC is the *information coefficient*, which is the correlation of the manager's forecast with the actual returns (how good the forecasts are); and BR is the breadth of the strategy (how many bets are taken). Breadth is the number of securities that can be traded and how frequently they can be traded. High information ratios are generated by a manager finding opportunities—and many of them—where she can forecast well. Grinold and Kahn (1999) state that "it is important to play often (high breadth, BR) and to play well (high IC)."

The fundamental law has been quite influential in active quantitative portfolio management because it offers a guideline as to how good asset managers have to be at forecasting, how many bets they have to make, or both to generate alpha. Suppose we require an information ratio of 0.5 and we make bets every quarter. A stock market timer trading just the aggregate market has narrow breadth because there are few assets—sometimes just stocks and bonds—to trade. Making only four bets a year, the stock market timer's bets have to be very accurate. She needs an IC of 0.25 to obtain an information ratio of $0.50 = 0.25 \times \sqrt{4}$. The cross-sectional strategies of value, size, and momentum (see chapter 7) have great breadth because hundreds of stocks can be traded. These strategies can have very low IC s, often just 2% to 5%, to be highly profitable. Stock selectors doing value, size, or momentum strategies making four hundred independent bets per year need only IC s of 0.025 to generate information ratios of $0.50 = 0.025 \times \sqrt{400}$. In short, you can be very talented at forecasting and make those bets really count, or you can have a very small edge but make a lot of bets. Both lead to high information ratios.

Grinold's fundamental law is derived under mean-variance utility, and so all the shortcomings of mean-variance utility apply (see chapter 3).³ In particular, by using mean-variance utility, the Grinold-Kahn framework ignores downside risk and other higher moment risk while assuming that all information is used optimally. A crucial assumption is that the forecasts are independent of each other. Nevertheless, the fundamental law is useful to frame the active management process. Alpha begins with raw information; we process that information into forecasts and then optimally and efficiently construct portfolios that balance

³The fundamental law is itself an approximation in the mean-variance framework. A better approximation to equation (10.5) is made by Hallerbach (2011). Extensions to the fundamental law allow time-varying signals for the forecasts (Ye (2008)), correlated forecasts (Buckle (2004)), and incorporate estimation risk (Zhou (2008)).